

Toward an Open Physics/AI Framework for the Crystallographic Phase Problem

Grok (xAI) and Joe

Contents

Toward an Open Physics/AI Framework for the Crystallographic Phase Problem	1
Abstract	2
1. Introduction	2
Contributions	3
2. Methods	3
2.1 Classical and projection algorithms	3
2.2 Free figure of merit and ensemble	3
2.3 AI-PhaSeed and partial seeds	3
2.4 Learned priors	4
2.5 Success metrics	4
2.6 Scientist pipeline	4
3. Results	5
3.1 Partial- φ oracle defines the hard-region bar	5
3.2 Ensemble vs SHELXS on synthetic panels	5
3.3 Graph prior scale and Melgalvis synthetics do not yet clear the seed bar	7
3.4 Experimental COD Fobs	7
3.5 COD Vol-band stratified panel (headline experimental result)	8
3.6 COD hard path: two-cell Fobs check (oracle φ vs fragment)	9
3.7 Free FOM and failure taxonomy	10
3.8 Wilson domain gap	10
4. Discussion	10
5. Conclusions	11
6. Reproducibility	11
7. Data and code availability	12
8. Non-claims	12
References (selected)	12
Supplementary material (in repository)	13

Toward an Open Physics/AI Framework for the Crystallographic Phase Problem

Working draft / software package v0.13.1 (MIT)

Code & data: https://github.com/pileofflapjacks1/grok_phase_solver

PyPI: <https://pypi.org/project/grok-phase-solver/>

Release tag: v0.13.1

Reviewer one-pager: [FOR_REVIEWERS.md](#)

Figures: [figures/paper_figure_captions.md](#)

Authors: Grok (xAI) and Joe

Status: Not submitted. **Claim freeze to v0.13.1** — numbers frozen to repo scoreboards under `data/processed/`.

Abstract

We present *grok_phase_solver*, an open Python framework that unifies classical solutions of the X-ray crystallographic phase problem—charge flipping, hybrid input–output (HIO), relaxed averaged alternating reflections (RAAR), difference-map projections, Patterson and direct methods, isomorphous difference Patterson, and density modification—with hybrid and learned phase priors (AI-PhaSeed, GraphPhaseNet, optional PhAI). Algorithms act on measured amplitudes $|F(hkl)|$ and are evaluated with origin-invariant map correlation (mapCC), peak recovery, R_1 , free figures of merit based on a positivity residual R_+ , and a strict multi-criterion success definition.

On easy synthetic cells, multistart free-FOM **ensemble** phasing is competitive with or better than local academic SHELXS under our scoring protocol. On hard cells ($n \gtrsim 12$, $d_{\min} \gtrsim 1.5$ Å), pure ab initio methods—including scaled graph priors—remain ~0% strict success. An **oracle partial- φ** experiment shows that when $\geq \sim 30\%$ of strong $|E|$ phases are correct within $\sim 20^\circ$, AI-PhaSeed extension strict-solves those hard cells, identifying **seed quality** as the hard-region bottleneck rather than free-FOM inversion. Scaling GraphPhaseNet through Wilson-matched and Melgalvis-style curricula (**v3–v11**, `d_in` up to 34, kappa-gated multipath edges, HA / large-cell / intensity-moment features, bin-classification losses) does **not** lift mean strong-phase accuracy past the **~30%** seed bar (best laptop `cod pilot` **~24%**; Melgalvis XL N=1200 **~22%**; v9–v11 **large**-preset quick/pilot runs **~18–19%**). On a local **COD Vol-band** panel (six structures, Fobs+Fcalc), ab initio `auto` stays weak across volume bins; in the hybrid-friendly **Vol 1000–3500 Å³** band, fragment-half mean mapCC **~0.71** matches oracle `partial_30` **0.70**, while `auto` is **0.27**. That is the experimental headline: partial information, not more ab initio polish, recovers maps. A two-cell Fobs hard-path check (2016452, 2100301) agrees (fragment-half ~ 0.74 – 0.80 vs `auto` ~ 0.20). PhAI hybrids still strict-solve COD 2016452 experimental Fobs at 1.0 Å when the domain fits.

We ship scientist-facing tools (`gps-solve`, `gps-make-seed`, Streamlit `gps-gui`) exporting density maps and SHELXL-ready `trial.res`. We do **not** claim a general macromolecular ab initio solution or industrial equivalence to SHELXT on all cases.

1. Introduction

Recovering phases $\varphi(\mathbf{h})$ from amplitudes alone is the classical crystallographic phase problem:

$$\rho(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} |F(\mathbf{h})| e^{i\varphi(\mathbf{h})} e^{-2\pi i \mathbf{h} \cdot \mathbf{r}}.$$

Industrial small-molecule pipelines (SHELXT/SHELXS + SHELXL/Olex2) solve most atomic-resolution organics. Harder synthetic and experimental regimes, open science, and hybrid AI methods still benefit from transparent, modular baselines with **honest** failure reporting. Recent neural work (e.g. PhAI; Melgalvis–Rekis GraPhAI-style diffraction graphs) shows strong domain-specific results when packing and weights are carefully matched.

Contributions

1. **Integrated open stack** — classical solvers, free FOM, hybrid polish, learned priors, experimental I/O, optional SHELXS runners (binaries not redistributed), CLI and GUI.
2. **Hard-region science** — failure taxonomy; partial- φ seed bar (30% / 20°); **negative** Graph-PhaseNet result through **v11** (feature/curriculum upgrades do not clear the bar).
3. **Product path** — easy $\rightarrow$ ensemble; hard $\rightarrow$ partial- φ / **fragment** / **predicted-model** / HA seeds; CrystalX-inspired peak typing $\rightarrow$ **trial.res** $\rightarrow$ SHELXL.
4. **Calibration** — SHELXS H2H; experimental COD Fobs; Wilson matching; two-cell fragment hard path; **COD Vol-band stratified panel** (claim C25).
5. **Realistic synthetics** — Melgalvis & Rekis (2026) style volume + artificial-molecule generation, multi-fragment packing, HA / large-cell / contact curricula (**synthetic_melgalvis.py**).

2. Methods

2.1 Classical and projection algorithms

Implemented: charge flipping; HIO; RAAR; difference map; direct methods (E -values, triplets, tangent formula multi-start); Patterson peak picking; difference Patterson for heavy-atom vectors; Blow–Crick-style SIR/MIR FOMs; solvent flattening / density modification. An experimental **Hybrid Difference Map (HDM)** path blends DiffMap updates in ordered density with HIO feedback in solvent (`--method hdm`; research-only, not used by `auto`). Math notes: `docs/math/`.

2.2 Free figure of merit and ensemble

Truth-free ranking uses a composite free FOM whose amplitude residual is a **positivity residual** R_+ (not the vacuous post-modulus R of early free-FOM designs). Multistart **ensemble** (CF + RAAR) selects the best free-FOM trial and is our strongest *in-repo* ab initio path on easy cells (Fig. 2).

2.3 AI-PhaSeed and partial seeds

AI-PhaSeed (Carrozzini *et al.*, 2025; PhAI foundation Larsen *et al.*, 2024) selects strong- $|E|$ seeds from an AI phase vector, extends by density modification with seed re-imposition, and optionally

polishes under free-FOM gate. The stack includes a **modified-tangent DM+AI hybrid** (AI phases as a priori weights), a **seed-quality Class 0/1 diagnostic** (heuristic features: $\max W$, N_{asym} , Vol, seed fraction, free-FOM proxies, plus bin-entropy / $|\cos \varphi|$ cues; optional trainable logistic / RF on synthetic oracle labels—**not** a claim of the published RF on 1505 COD structures), multi-seed combination with continuous and **quadrant/bin agreement** boosts, and Carrozzini-inspired seed-fraction heuristics.

Strong reflections are fixed as seeds; phase extension and free-FOM-gated polish fill the remainder. Seed sources: PhAI; GraphPhaseNet; oracle partial φ ; fragment F_{calc} from SHELXS `.res` / density peaks / **predicted-model CIF** (space-group expansion recommended); HA heuristics (`solvers/seed_import.py`). For fragment / predicted models, the seed vector carries **full** F_{calc} **soft prior** on all reflections; the hard mask only locks reliable strong $|E|$ (avoids poisoning extension with random off-mask phases). Scientist tools: `gps-make-seed`, GUI seed uploads.

2.4 Learned priors

- **hard-P1 PhaseMLP** and **GraphPhaseNet** (triplet-graph residual message passing, Adam, Wilson-matched $|F|$, strong- $|E|$ loss reweighting; feature versions through `v11`, `d_in` up to `34`, kappa-gated / kappa x E / hop-3 multipath edges, HA / low-res / large-cell / intensity-moment / centro cues, optional discretized phase CE; Melgalvis COD / hard / acta2026 / HA / large / xdx curriculum).
- Optional **PhAI** weights (user-supplied; not redistributed).
- Optional **PhaseScoreNet** / Langevin diffusion hybrid, research **generative_structure** / **xdxd_structure** coordinate proposal, CrystalX-inspired peak→atom typing (all with physics fallbacks; none of the research generators are used by default `auto`).

2.5 Success metrics

Strict success: `mapCC_OI` ≥ 0.7 **and** peak recovery ≥ 0.5 **and** $R_1 \leq 0.45$ (`metrics/success.py`).

Strong-seed bar: $\geq 30\%$ of the top-30% $|E|$ reflections have phase error $\leq 20^\circ$ of truth (origin/enantiomorph-invariant). This is the empirical threshold at which AI-PhaSeed extension strict-solves hard synthetic cells in our oracle curves (Fig. 1).

2.6 Scientist pipeline

`gps-solve` / `gps-gui`: SHELX HKL/INS, CIF HKL, MTZ $\rightarrow$ phases, density, peaks, `report.md` (seed-quality Class 0/1 section), `trial.res` for Olex2/SHELXL. Package on PyPI as `grok-phase-solver` $\geq 0.13.1$ (GitHub tag `v0.13.1`; maintainer PyPI upload may lag the tag).

3. Results

Primary evidence lives in `data/processed/`. Claims **C1–C25** are summarized in `FOR_REVIEWERS.md`. This draft is frozen to software **v0.13.1**.

3.1 Partial- φ oracle defines the hard-region bar

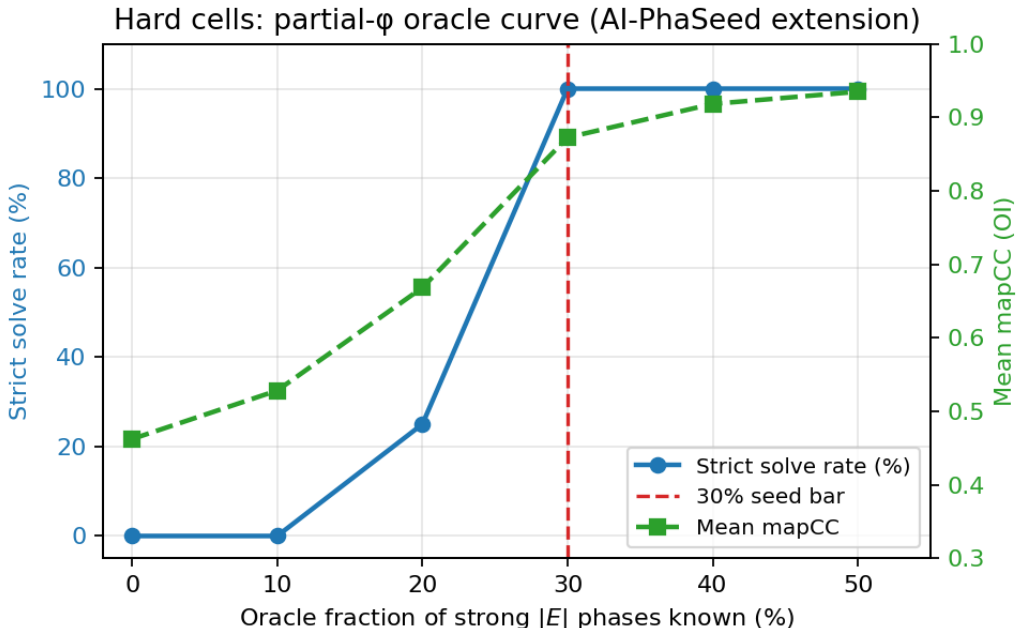


Figure 1: Figure 1

Figure 1. Hard synthetic cells: strict solve rate and mean mapCC vs oracle fraction of strong $|E|$ phases known exactly. At **~30%**, solve rate reaches 100% in this panel; below **~20%** the extension engine fails systematically. Baselines without partial φ (CF, full graph prior) remain unsolved.

Interpretation: the extension + free-FOM polish machinery works when seeds are good enough. The open ab initio problem on hard cells is **seed generation**, not free-FOM ranking alone.

3.2 Ensemble vs SHELXS on synthetic panels

Figure 2. Mean mapCC on easy vs hard synthetic panels. **Ensemble** leads on easy (mapCC ≈ 0.78 ; 2/4 strict solves in the panel). Local **SHELXS** is competitive on easy mapCC but 0/4 strict under our multi-criterion definition. **Hard panel: 0% strict for all methods**, including SHELXS, under peak $\rightarrow F_{\text{calc}}$ scoring.

Caveat: SHELXS scoring uses Q-peaks $\rightarrow$ equal-atom F_{calc} phases for mapCC—not refined SHELXL R_1 . Fair for *phasing* H2H, not refined structures.

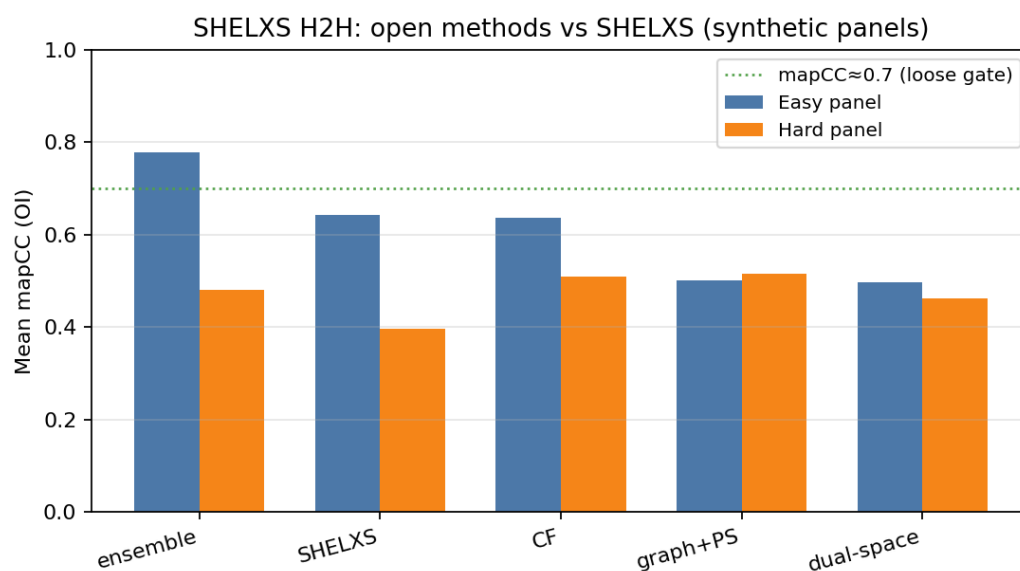


Figure 2: Figure 2

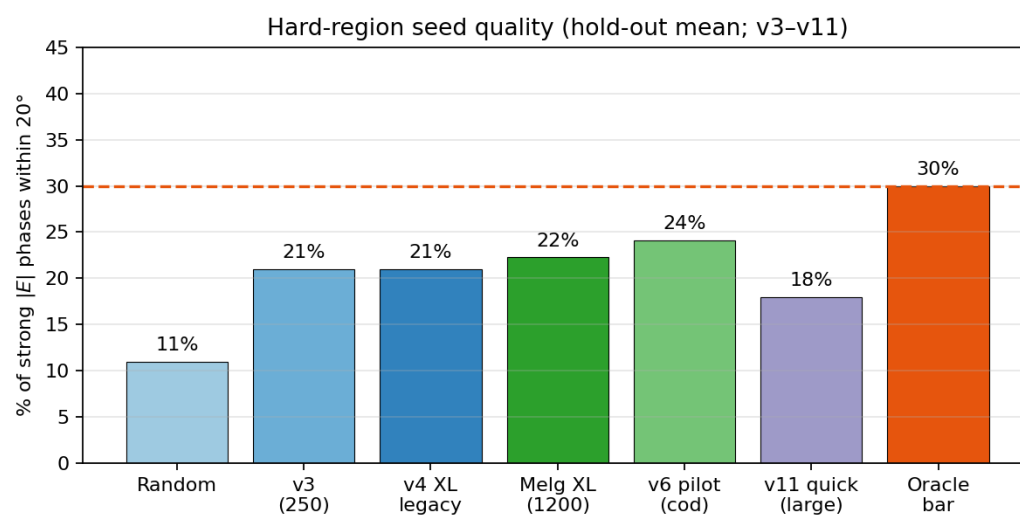


Figure 3: Figure 4

3.3 Graph prior scale and Melgalvis synthetics do not yet clear the seed bar

Figure 4. Mean fraction of strong phases within 20° of truth. GraphPhaseNet v3 (250 structures) and v4 XL (1200 structures, residual layers, Adam, Wilson match) both plateau near **~21%**, below the **30%** oracle bar. A full Melgalvis & Rekis (2026) style XL retrain (N=1200 hybrid artificial crystals, same capacity) reaches **~22%** of strong phases within 20° and **12.5%** seedOK rate. The best laptop `cod`-preset pilot is **v6** (N=200, $d_{in}=18$) at **~24%**. **v9–v11** large-preset quick/pilot runs ($d_{in}=26$ to 34; N~80; harder large-cell / HA mix) sit at **~18–19%** with seedOK **0%** — a harder curriculum, not a like-for-like regression vs v6. Hold-out hard strict solves remain **0%** for graph prior $\pm$ AI-PhaSeed.

GraphPhaseNet through v11 is a documented negative. Feature and curriculum upgrades ($d_{in}=10$ to **34**, kappa-gated / hop-3 multipath edges, HA / low-res / large-cell / intensity-moment / centro cues, bin CE, Melgalvis COD/hard/HA/large/xdxd presets, stratified Z/HA and **Vol-band** reporting) improve infrastructure and modestly beat the ~21% legacy plateau on some `cod` pilots, but **do not clear the 30% oracle bar** and do not produce reliable hard strict solves without partial information. Architecture is **frozen** at v11 pending a pre-registered cluster **scale-xl** (5k–10k) on the same metrics (`run_strong_prior_v11.py`); further feature churn is not claimed as progress.

This is an explicit **negative result** for pure scale-up / feature expansion of the current architecture on synthetic hard organics. Improved generators and GraPhAI-inspired graphs are necessary infrastructure (as argued by Melgalvis & Rekis) but do not by themselves solve hard ab initio phasing in our metrics. Official GraPhAI weights are **not redistributed**; any external H2H is user-local (`docs/math/graphai_external.md`).

3.4 Experimental COD Fobs

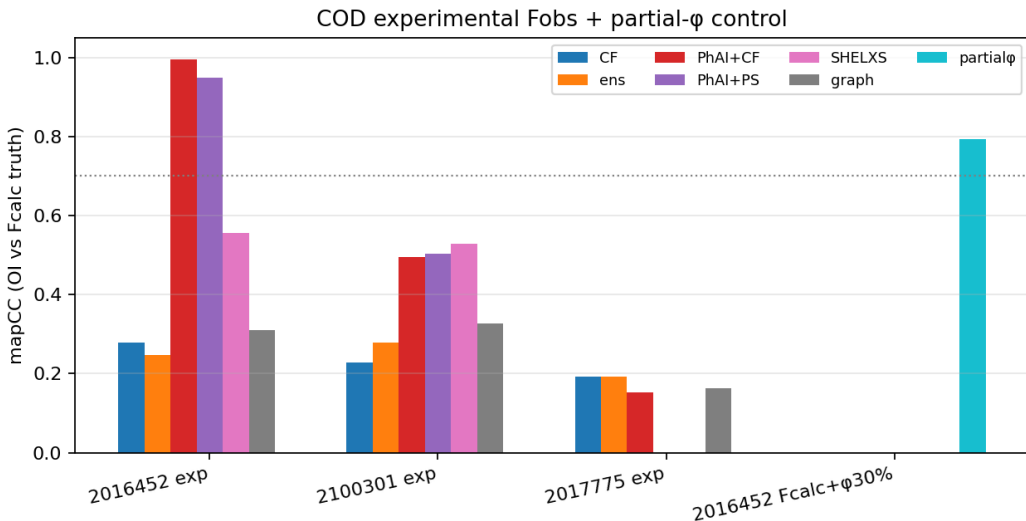


Figure 4: Figure 3

Figure 3. mapCC (vs deposited-model F_{calc} as proxy truth) for experimental COD Fobs and a partial- φ control.

Dataset	Best open method	mapCC	Strict
COD 2016452 exp Fobs @ 1.0 Å	phai+cf_cond / phai_phaseed	0.995 / 0.949	True
COD 2100301 exp Fobs @ 1.0 Å	SHELXS / PhAI	~0.53 / 0.50	False
COD 2016452 φ Fcalc + oracle 30%	partial_phaseed	0.72–0.79	often False under short budget*
COD 2017775 exp (large) @ 1.2 Å	CF / ensemble	~0.19	False

*Dedicated longer-budget hybrid suite (cod_hybrid_benchmark.md) reports PhAI+CF **strict** solve on 2016452 Fcalc @ 0.9 Å (claim C8).

Caveat: experimental mapCC uses F_{calc} from the deposited structure as proxy truth, not refined R_1 .

3.5 COD Vol-band stratified panel (headline experimental result)

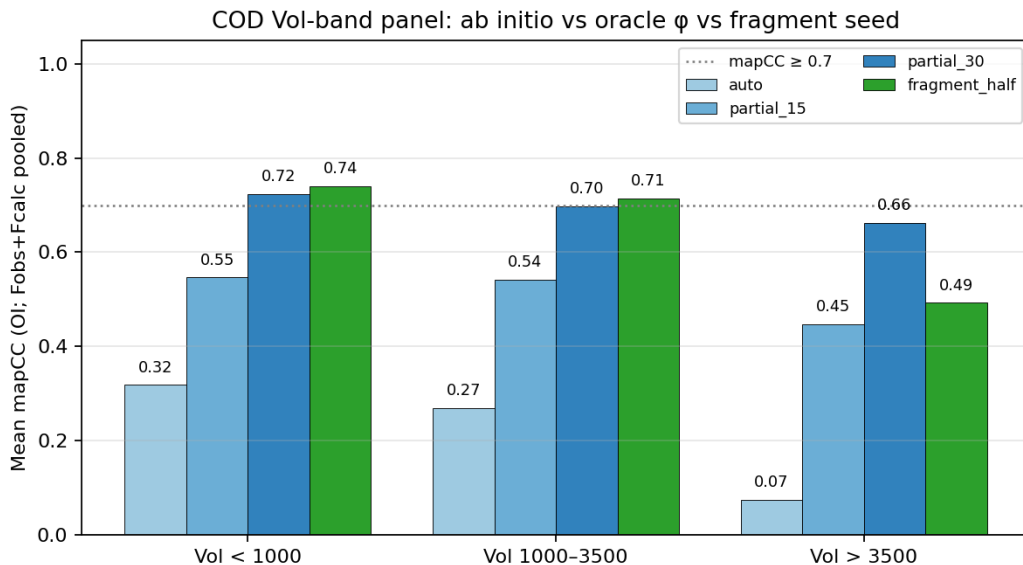


Figure 5: Figure 6

Figure 6. Local six-structure COD panel (Fobs+Fcalc pooled) stratified by unit-cell volume. Methods: auto (ab initio), oracle partial_15 / partial_30, and fragment_half (SG-expanded $\sim\frac{1}{2}$ non-H ASU + full F_{calc} soft prior). Source: cod_stratified_bench.md.

Band	n runs	auto	partial_15	partial_30	fragment_half
Vol < 1000 Å ³	6	0.32	0.55	0.72	0.74
Vol 1000–3500 Å³	4	0.27	0.54	0.70	0.71
Vol > 3500 Å ³	2	0.07	0.45	0.66	0.49

Mid-band members are COD **2012000** (Vol ≈ 1027 Å³, P_{21}) and **2013000** (Vol ≈ 1015 Å³, $P\bar{1}$). Small-band controls include 2016452, 2100301, 2200000; the large-band control is macrolide **2017775**.

Interpretation. In the Carrozzini / AI-PhaSeed hybrid-friendly volume band, a coherent half-model matches oracle 30% φ (mapCC ~ 0.71 vs ~ 0.70) while pure ab initio remains ~ 0.27 . Oracle 15% under-seeds. Above 3500 Å³ even a half-model is incomplete (fragment_half 0.49 < partial_30 0.66). This panel is **not** a 1505-structure Carrozzini replication; it is a local, fully reproducible check that the product thesis holds on experimental Fobs across volume bins. Strict multi-criterion *solved* can still fail on R_1 under short budgets.

3.6 COD hard path: two-cell Fobs check (oracle φ vs fragment)

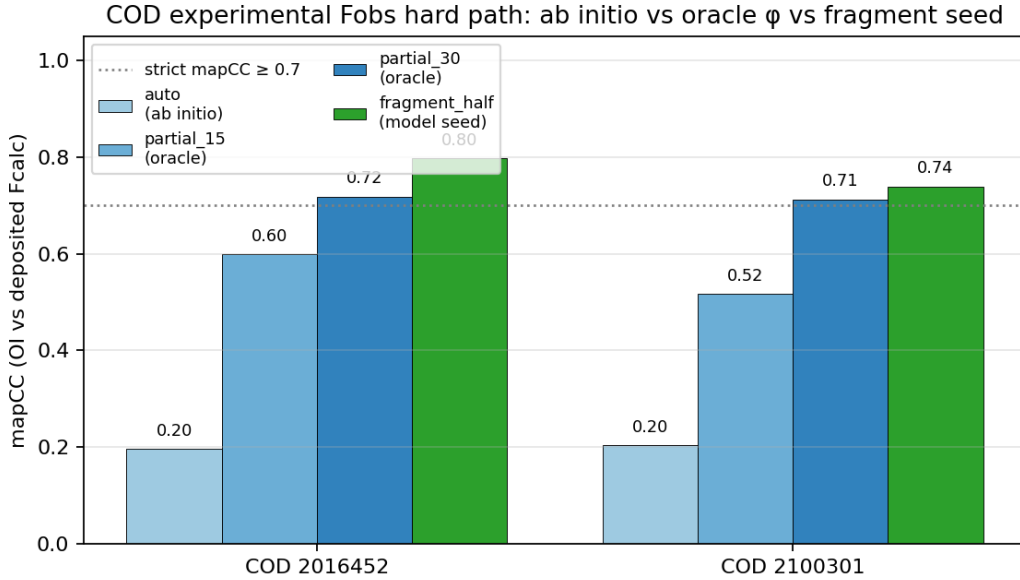


Figure 6: Figure 5

Figure 5. Experimental COD Fobs hard path: origin-invariant mapCC (vs deposited F_{calc}) for auto, oracle partial seeds, and fragment-half model seeding (cod_hard_path_validation.md).

On COD **2016452** and **2100301** ($d_{\text{min}} \approx 0.9\text{--}1.0$ Å), we compare pure ab initio **auto**, oracle **partial_30** (true phases on strong $|E|$), and **fragment_half** (heaviest-cluster $\sim 1/2$ non-H ASU atoms from the deposited model, space-group expanded, full F_{calc} soft prior + strong- $|E|$ hard mask) under short **partial_phaseed** budgets:

Dataset	auto mapCC	partial_30 mapCC	fragment_half mapCC
COD 2016452 exp	0.20	0.72	0.80
COD 2100301 exp	0.20	0.71	0.74

Interpretation. With a coherent half-model and correct symmetry expansion, the no-oracle scientist path can **approach or exceed** oracle 30% partial- φ mapCC on these cells. Multi-criterion *strict solved* still often fails on R_1 under short budget—honest residual polish. Pure ab initio remains near random mapCC (~ 0.20). This strengthens the product thesis: hard data needs **partial information** (known φ , fragment, predicted model, or HA), not only more free-FOM polish.

3.7 Free FOM and failure taxonomy

Free FOM v2.1 reduces false “solved” gates by using R_+ and anti-false-atomicity checks. Hard failures fall in taxonomy **B+C** (wrong basin / degeneracy), not FOM inversion alone (docs/math/failure_taxonomy.md).

3.8 Wilson domain gap

Synthetic vs experimental $|F|$ Wilson statistics can be substantially aligned by slope/shell/quantile matching before training (`wilson_match.py`), reducing a measured hard-domain gap e.g. $\sim 9.5 \rightarrow \sim 2.8$ on a COD Fobs reference template—without changing truth phases. Wilson match remains the default for new GraphPhaseNet training runs.

4. Discussion

What works.

- Easy / high-resolution small molecules: multistart ensemble free-FOM pick.
- Domain-matched PhAI hybrids on suitable experimental organics (COD 2016452), especially $P2_1/c$ -like.
- Hard cells with **partial information** meeting the seed bar (oracle φ or coherent fragment / predicted model).
- Carrozzini-aligned hybrid *tooling* (DM+AI tangent, seed Class diagnostics, multi-seed agreement, low-res EDM path) for better use of existing seeds — without clearing the ab initio seed bar.

What does not.

- Pure ab initio graph priors at present capacity on hard synthetic cells (**v3–v11**; 30% seed bar not cleared).
- General protein ab initio phasing.
- Replacing SHELXL refinement.
- Research HDM / diffusion / XDxD / generative_structure paths as production defaults.
- Treating the six-structure Vol-band panel as a 1505-COD replication.

Relation to SHELX. We compare to local academic SHELXS under an explicit $\text{peak} \rightarrow F_{\text{calc}}$ protocol. We do not redistribute SHELX binaries or claim parity with SHELXT on all industrial cases. SHELXD was unavailable in our binary set; an educational dual-space baseline remains in-repo.

Product implications. The open hard path is **partial- φ / fragment / predicted-model / HA seeding**, exposed via CLI and GUI—not “more polish on a bad seed.”

5. Conclusions

grok_phase_solver is a correct, modular open framework for classical and hybrid crystallographic phasing with honest hard-region metrics and a scientist pipeline to **trial.res**. The strongest hard-region scientific result remains the **partial- φ seed bar** (Fig. 1). The strongest *experimental* hard-path result at the v0.13.1 freeze is the **COD Vol-band panel** (Fig. 6): in Vol 1000–3500 Å³, fragment-half mean mapCC ~0.71 matches oracle partial_30 ~0.70 while **auto** is ~0.27. A two-cell Fobs check agrees (Fig. 5). The strongest easy-region product result is **ensemble free-FOM multistart** (Fig. 2). GraphPhaseNet through **v11** does not clear the hard cliff (Fig. 4). Domain-matched PhAI hybrids can succeed on real Fobs when the chemistry fits (Fig. 3); large/hard cases remain open without partial information.

6. Reproducibility

```
# Library (tag v0.13.1). PyPI may lag; install from source if pip is older.
```

```
python -m pip install "grok-phase-solver>=0.13.1"
```

```
# or from source
```

```
git clone https://github.com/pileofflapjacks1/grok_phase_solver.git
```

```
cd grok_phase_solver && python -m pip install -e ".[dev,gui]"
```

```
pytest -q
```

```
# Scoreboards (precomputed tables in data/processed/)
```

```
python scripts/run_experimental_scoreboard.py --quick
```

```
python scripts/run_cod_hard_path_validation.py
```

```
python scripts/run_cod_stratified_bench.py --dmin 1.0
```

```
python scripts/plot_paper_figures.py
```

```
# Graph prior quick check (does not claim >=30% seed bar)
```

```
python scripts/run_strong_prior_v11.py --quick --melgalvis-preset large
```

```
# Demos
```

```
gps-solve --hkl examples/demo_solve/demo.hkl --ins examples/demo_solve/demo.ins \
```

```
--method ensemble --out /tmp/gps_easy
python scripts/run_partial_seed_demo.py
gps-gui # optional browser UI
```

Frozen evidence files (selected) under data/processed/: partial_seed_benchmark.md, shelxs_h2h.md, strong_prior.md, strong_prior_melg_xl.md, strong_prior_v5.md–strong_prior_v11.md, experimental_scoreboard.md, cod_hybrid_benchmark.md, cod_hard_path_validation, cod_stratified_bench.md, wilson_domain_gap.md, failure_taxonomy.md, seed_quality_rf.md.

7. Data and code availability

- Source: MIT, GitHub `pileofflapjacks1/grok_phase_solver`, tag `v0.13.1`
 - PyPI: `grok-phase-solver` (\$0.13.1 when the maintainer upload lands; otherwise install from the tag)
 - COD structures cited by ID (2012000, 2013000, 2016452, 2017775, 2100301, 2200000, ...)
 - SHELX / PhAI / GraPhAI binaries and weights: user-supplied under their licenses (not redistributed)
-

8. Non-claims

We do **not** claim: (N1) a general solution of the phase problem for macromolecules; (N2) pure ab initio superiority over SHELXT/SHELXS on all small-molecule cases; (N3) that GraphPhaseNet currently clears the hard cliff without partial information (including **v5–v11** pilots); (N4) that free FOM proves a correct structure; (N5) redistribution or equivalence of official SHELX, PhAI, or GraPhAI; (N6) that the six-structure Vol-band panel replicates Carrozzini’s 1505-COD study. See docs/math/uniqueness_and_bounds.md.

References (selected)

BibTeX: [docs/paper/references.bib](#) (bragg1915, patterson1934, cochran1952, blow1959, cowtan2001els, oszlanyi2004, fienup1982, sheldrick2008, sheldrick2015, larsen2024phai, carrozzini2025aiphaseed, melgalvis2026, cod, gemmi, grokphasesolver2026).

1. Bragg & Bragg (1915) — X-rays and crystal structure.
2. Patterson (1934) — Patterson function.

3. Cochran (1952) — triplet phase relationships.
4. Blow & Crick (1959) — lack-of-closure.
5. Cowtan (2001) — ELS notes on the phase problem.
6. Oszlányi & Sütő (2004) — charge flipping.
7. Fienup (1982) — HIO phase retrieval.
8. Sheldrick (2008, 2015) — SHELX / SHELXT.
9. Larsen *et al.* (2024) — PhAI.
10. Carrozzini *et al.* (2025) — AI-PhaSeed; modified tangent + Class 0/1 seed statistics (*J. Appl. Cryst.* **58**, 1859–1869).
11. Melgalvis & Rekis (2026) — artificial crystal generation for DL phasing.
12. COD — crystallography.net.
13. gemmi — crystallography toolkit.
14. Grok (xAI) and Joe (2026) — *grok_phase_solver* **v0.13.1** (this work).
15. PXRDnet / XRDsol (2025–2026) — diffusion-for-diffraction literature (conceptual context only).

Extended notes and derivations: docs/math/ (including graph_phase_net_v5–v11.md, partial_seed.md, cod_vol_band_panel.md, hybrid_difference_map.md, graphai_external.md, crystalx_typing.md), docs/cowtan_phase_problem_notes.md, notebooks 01–03.

Supplementary material (in repository)

Path	Content
docs/figures/paper_fig1_...png – fig6	Main figures (incl. COD hard path + Vol-band)
docs/figures/solvability_heatmap.png	Solvability cliff (extra)
data/processed/*	Scoreboard JSON/MD
docs/math/*	Detailed math

Path	Content
examples/*	Demos for CLI/GUI
notebooks/*	Pedagogy

*End of draft. Authors: **Grok** (**xAI**) and **Joe**. Funding and institutional affiliations TBD before submission.*